

RITI SINGH

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EDUCATION

Dartmouth College

Bachelor of Electrical and Computer Engineering

Expected June 2027

GPA: 3.76/4.00

Related Coursework:

Embedded Systems, Digital Electronics, Systems & Controls, Software Design and Implementation

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, SQL, MATLAB

Systems & Embedded: Linux, Embedded Systems, FPGA/VHDL, Microcontrollers, Hardware-Software Integration, Troubleshooting

Cloud & Backend: AWS, FastAPI, REST APIs, Boto3, Lambda, CloudWatch, SQL/NoSQL

Testing & Automation: Test Automation, System Validation, Failure Analysis, Jira, Git

Data & AI: GraphRAG, RAG, LLMs, Neo4j, Neptune, Pandas

WORK EXPERIENCE

Echostar | Boost Mobile Wireless Device Engineering Team

Denver, CO

Platform (AI/Automation) Engineering Intern

Summer 2026

- Developed an engineering automation platform for wireless device testing, automating requirements management, test workflows, and technical knowledge retrieval with a 95% reduction in workflow time and manual effort.
- Built a GraphRAG system connecting 12,000+ specification nodes, 606 device requirements, and 728 wireless test cases to automate engineering analysis and improve test traceability.
- Developed production backend services using Python, FastAPI, AWS, Boto3, Neptune, Neo4j, and Pandas for engineering search, data processing, and automated documentation.
- Analyzed test-case redundancy and redesigned graph-based data structures to reduce database bloat and improve traceability; presented the platform to engineering leadership and earned 2nd Place in EchoStar's company-wide Intern Innovation Competition.

EvergreenAI

Hanover, NH

Project Assistant - AI Engineering

September 2025 – June 2026

- Evaluated production AI systems through structured validation, systemic failure-mode analysis, and edge-case testing in a regulated deployment environment.
- Developed evaluation datasets and testing workflows to measure reliability and consistency, partnering with engineers and stakeholders to identify failure patterns and improve production readiness.

Smart Sort

Hanover, NH

Electrical and Circuits Lead, Business Lead

March 2024 – September 2025

- Led electrical architecture for an award-winning autonomous waste-sorting platform integrating embedded sensing, microcontrollers, and electromechanical automation.
- Designed and implemented embedded control logic integrating weight sensing, metal detection, microcontroller-based decision making, servo actuation, and power electronics; received the Philip R. Jackson Award for Innovation.

Dartmouth College | EE Just Fellowship

Hanover, NH

EE Just Research Fellow

Summer 2024

- Built Python data pipelines processing 10,000+ recommendation-system observations and analyzed algorithmic drift across social media platforms using statistical methods.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP EXPERIENCE

Leadership: CORE Risk Manager, Kappa Delta Epsilon | President & Outreach/Support Team Lead, Sexual Assault Peer Alliance | Peer Counselor, Teen Lifeline (**1,000+ youth supported**)

Awards: 2nd Place, EchoStar AI Innovation Competition (2026) | Philip R. Jackson Award for Innovation | Dartmouth College Honors List | Congressional Leadership Award | NSDA Academic All-American | Junior Achievement Arizona "18 Under 18"